

DEFENSE MEDICAL COLLEGE

HISTOLOGY Progressive assessment II for Year I Medical Students

Time allowed: 2:00 hours.

Name: _____

Date: April 30, 2014

For each of the following **Multiple Choice Questions** choose the one **BEST** appropriate answer and mark 'X' in the appropriate box located against the corresponding question number and letter in the separate answer sheet provided.

- _____1. Which of the skin's many functions are partly or completely provided by components of the **dermis**?
- a) protection from dehydration b) regulation of blood pressure c) synthesis of melanin
d) protection from ultraviolet radiation e) synthesis of keratin
- _____2. Which connective tissue layer lies **closest to the epidermis**:
- a. reticular layer of dermis. b. papillary layer of dermis. c. hypodermis. d. lamina propria. e. mucosa.
- _____3. The majority of cells comprising the **epidermis** are called:
- a. keratinocytes b. melanocytes. c. Langerhans cells. d. Merkel cells. e. fibroblasts.
- _____4. Which of the following is **NOT** an "**epidermal appendage**" (i.e., NOT comprised of epithelial tissue)?
- a.sebaceous gland. b. hair follicle. c.duct of sweat gland.
d. secretory portion of sweat gland. e. arrector pili.
- _____5. Which cell type in the dermis is most closely associated with stimulating an **inflammatory response**, by secreting **histamine** and other chemical signals?
- a) fibroblast. b) Langerhans cells. c) macrophage. d) lymphocyte. e) mast cell.
- _____6. Which of the following **epidermal layers** is most likely to contain both **desmosomes** & **hemidesmosomes**?
- a) stratum basale b) stratum corneum c) stratum granulosum
d) stratum lucidum e) stratum spinosum
- _____7. **Arrector pili muscles**,
- a) are composed of bundles of myoepithelial cells. b) help push sebum onto the skin surface.
c) have no effect on hair position in humans. d) contain capillary networks that nourish growing hairs.
e) attach directly to the external root sheath by desmosomes.
- _____8. The term **mucosa** (mucous membrane) means:
- a. a nonkeratinized stratified squamous epithelium.
b. an epithelium which contains mucus-secreting goblet cells.
c. the moist surface of any hollow organ that communicates with the outside world.

d. a body surface onto which mucous glands secrete their product. e. the lining of the GI tract.

_____9. Which of the following is lined by a **mucosa**?

- a. Peritoneal cavity
- b. Pericardial cavity
- c. Pleural cavity
- d. Alimentary canal
- e. All of the above are lined by a mucosa

_____10. The **stomach**:

- a) Has its muscularis externa made up of skeletal muscle fibres.
- b) Has its gastric glands best developed in the pylorus than the other regions.
- c) Has chief cells that are intensely eosinophilic.
- d) Has parietal cells in its glands that secrete HCl.
- e) Is able to withstand the acidic environment because it is lined by stratified squamous keratinized epithelium.

_____11. Concerning the **teeth and associated tissues**, which statement is **true**?

- a) Dentin is more calcified than enamel.
- b) Pulp is made up of dense connective tissue.
- c) Enamel is replaced in the root by cementum.
- d) Odontoblasts make enamel.
- e) Periodontal membrane is loose connective tissue.

_____12. Which of the following statements is **correct** regarding the **intestine**?

- a) Large intestine possesses branched alveolar glands.
- b) The striated border of the small intestine is made by microvilli.
- c) Small intestine has taenia coli.
- d) Large intestine contains villi.
- e) None of the above.

_____13. **Mucosal glands of the pylorus**,

- a) resemble those in the body of the stomach.
- b) secrete more pepsinogen than do fundic glands.
- c) have shallower pits than those in the fundic region.
- d) secrete mainly mucus.
- e) contain abundant parietal cells in their basal half.

_____14. **Permanent circular folds (plicae circulares)** are characteristic structural feature of the walls of which of the following organs?

- a) anal canal
- b) colon
- c) cecum
- d) small intestine
- e) stomach

_____15. Which of the following organs characteristically contains the most **goblet cells** in the mucosa covering its villi?

- a) duodenum
- b) ileum
- c) colon
- d) cecum
- e) stomach

_____16. **Brunner's glands**,

- a) are characteristic components of the jejunal wall.
- b) lie in the submucosal layer.
- c) empty their secretions into the lacteals.
- d) are composed of collections of goblet cells.
- e) produce serous secretion that is rich in digestive enzymes.

_____17. **Kupffer cells**,

- a) are nonphagocytic
- b) contain lysosomes
- c) are derived from circulating neutrophils
- d) resemble hepatocytes in HE-stained sections
- e) border on the bile canaliculi

_____18. Which of the following glands lacks **serous acini** but contains **serous demilunes**?

- a) liver
- b) parotid gland
- c) pancreas
- d) sublingual gland
- e) submandibular gland

_____19. The whitish tips of **filiform papillae** consist of:

- a. keratinized epithelial tissue. b. collagen. c. mucous. d. taste buds. e. goblet cells.

_____20. The **myenteric (Auerbach's) plexus** is located:

- a. just beneath the epithelium of the mucosa, in lamina propria. b. in submucosa.
c. within lamina propria, adjacent to the muscularis mucosae. d. in serosa and/or adventitia.
e. between the layers of smooth muscle in the muscularis (externa).

_____21. In the digestive system, **lymph nodules** occur:

- a. concentrated in the tonsils. b. concentrated in the appendix. c. concentrated in Peyer's patches.
d. scattered in lamina propria throughout the GI tract. e. all of the above.

_____22. **Submucosal glands** occur in the:

- a. duodenum and esophagus. b. Stomach and ileum. c. jejunum and colon.
d. appendix and anal canal. e. All of them.

_____23. Select the **incorrect** statement with regard to the **GI tract**.

- a. Folds in the mucosa of the stomach are called rugae.
b. Lymphatic vessels within intestinal villi, which transport lipids to the circulatory system, are called lacteals.
c. The basal surface of hepatocytes is bathed by blood plasma as it flows freely through the space of Disse.
d. The principal secretory product for cells of Brunner's (duodenal) glands is acid mucus.
e. The principal secretory product for the Paneth cells clustered at the ends of intestinal crypts is lysosomal enzymes.

_____24. Which of the following regions of the **GI tract** is characterized by **stratified squamous epithelium**?

- a. duodenum b. jejunum c. ileum d. colon e. anal canal

_____25. In which of the following regions of the **GI tract** does the mucosa most closely resemble that of the **colon**, but with more **lymphoid tissue**?

- a. stomach b. duodenum c. jejunum d. ileum e. appendix

_____26. A functional unit of **liver** tissue served by a single terminal branch of the hepatic artery, including all the hepatic cords leading toward central veins on both sides of this arteriole, is called a:

- a. liver acinus. b. portal triad, tract or canal. c. classical or hepatic lobule.
d. sinusoid. e. biliary tract.

_____27. A **hepatic cord** is:

- a. another name for a branch of the portal vein. b. the tough connective tissue surrounding a portal triad.
c. a row of adjacent hepatocytes within a liver lobule. d. a branch of the bile duct. e. a nerve within the liver.

_____28. Choose the best description of the **colon**.

- a. Mucosa with villi and short tubular invaginations; epithelium with many goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.
b. Mucosa with short tubular invaginations but no villi; epithelium consisting entirely of goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.
c. Mucosa with villi and short tubular invaginations; epithelium consisting entirely of goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.
d. Mucosa thick, with many tubular glands consisting of several distinct secretory cell types; epithelium with many goblet cells; lamina propria inconspicuous; submucosa unspecialized.

e. Mucosa with short tubular invaginations but no villi; epithelium with many goblet cells; lamina propria with occasional lymph nodules; submucosa unspecialized.

———29. **Kupffer cells** are found most commonly within the:

- a. gall bladder. b. portal canals. c. hepatic sinusoids. d. central vein. e. bile ducts.

———30. **Hepatocyte** function includes all of the following **EXCEPT**:

- a. drug detoxification. b. protein synthesis. c. gluconeogenesis.
d. urea formation by deamination of amino acids. e. breakdown of hemoglobin.

———31. **Fat-storing stellate cells (Ito cells)** in the space of Disse are noted for storing:

- a. glycogen. b. glucose. c. albumin. d. vitamin A. e. all of the above.

———32. Like the small intestine, the **gall bladder**:

- a. is lined by columnar cells with a brush border of microvilli. b. has goblet cells in the mucosal epithelium.
c. has crypts with Paneth cells at the bottom. d. has villi with lacteals in the core.
e. has a distinct muscularis mucosa, and is further surrounded by two distinct layers of circular and longitudinal muscle.

———33. Which of the following is characteristic of the **proximal** but **not** of the **distal convoluted tubules**?

- a) convoluted portion lies in the renal cortex b) epithelial lining is similar to that of the loop of Henle
c) exhibits a brush border d) lining cells are basophilic
e) secretes glucose and amino acids into the filtrate

———34. Which of the following is the **modified smooth muscle** cell responsible for the **secretion of renin**?

- a) juxtaglomerular cell b) mesangial cell c) podocyte
d) polkissen (extraglomerular mesangial cell) e) macula densa cell

———35. Which of the following is **not** a **function of the kidney**?

- a. Erythropoietin production b. Vitamin D modification c. Acid base balance
d. Aldosterone production e. Renin production

———36. What vessels arise from the **efferent arteriole**?

- a. Afferent arteriole b. Efferent arteriole c. Capillary tuft d. Peritubular capillaries e. Vasa recta

———37. Which of the following is **not** the function of **mesangial cells**?

- a. synthesis of extracellular matrix. b. endocytosis and contractility.
c. maintenance of acid-base balance in the blood. d. structural support to the glomerulus.
e. increase blood flow and the effective surface area available for filtration.

———38. The tip of each **papilla**, perforated by the openings of **collecting ducts**, is called the:

- a. medullary ray. b. capsule. c. column of Bertin. d. area cribrosa. e. hilus.

———39. **Filtration slits** are located between adjacent:

- a. layers of the glomerular basement membrane. b. capillary fenestrations.
c. podocyte foot processes. d. mesangial cells. e. capillary endothelial cells.

———40. Epithelial cells lining the thin limb of the **loop of Henle**:

- a. have an apical brush border of microvilli for active absorption from the lumen.

- b. contain abundant ribosomes and rough endoplasmic reticulum for extensive protein synthesis.
- c. contain numerous basally located secretory vesicles.
- d. are simple squamous.
- e. all of the above.

———41. The **macula densa** consists of:

- a. closely-packed nuclei in the epithelium of the distal convoluted tubule.
- b. closely-packed nuclei in the epithelium of the proximal convoluted tubule.
- c. densely packed collagen fibers in the juxtaglomerular interstitium.
- d. modified smooth muscle cells.
- e. extraglomerular mesangial cells.

———42. **Transitional epithelium** lines all of the following **EXCEPT** the:

- a. renal pelvis.
- b. ureter.
- c. bladder.
- d. proximal urethra.
- e. Bowman's capsule.

———43. In the **urinary system**, which statement is **false**?

- a. The ureter is lined by transitional epithelium and it does not contain lymphatics.
- b. Polyploidy and binucleated cells are common among cells of the transitional epithelium.
- c. Distal convoluted tubule is longer and more convoluted than proximal convoluted tubule.
- d. Calyces are lined by transitional epithelium.
- e. Aldosterone acts on the distal convoluted duct.

———44. Concerning the **chromophobes of pars distalis of adenohypophysis**, which statement is **false**?

- a. may represent stem cells that have not yet differentiated into hormone-producing cells.
- b. maintain favorable fluid and electrolyte environment for the chromophil secretory cells.
- c. may be acidophils or basophils that are degranulated and thereby are depleted of hormone.
- d. do not stain intensely.
- e. may have minimal or no hormonal content.

———45. Which of the following hormones is carried to its target(s) by the **hypophyseal portal system**?

- a) ADH
- b) ACTH
- c) FSH
- d) TRH
- e) TSH

———46. Which of the following hormones is **most** responsible for **increasing osteoclast number and activity**?

- a) ADH
- b) calcitonin
- c) estrogen
- d) parathyroid hormone
- e) testosterone

———47. Which statement is **false** with regard to the **epiphysis cerebri {pineal body}**?

- a. Melatonin may "delay" puberty through antigonadotropic effects.
- b. Its secretory activity is stimulated by darkness and inhibited by light.
- c. Brain sand or corpora arenacea increase in size and number with increasing age.
- d. is surrounded by pia mater, which functions as its capsule.
- e. None of the above.

———48. Which statement is **true** with regard to the **endocrine system**?

- a. Myxedema is considered to be an autoimmune disease caused by hyperthyroidism.
- b. Calcitonin, secreted by the parathyroid gland, stimulates calcium resorption in the bones.
- c. The small and most numerous oxyphil cells of the parathyroid glands stain strongly basophilic.
- d. Both zona fasciculata and zona reticularis of the suprarenal gland depend on ACTH to sustain their function and survival.
- e. The cells of the zona fasciculata of the suprarenal gland are called chromaffin cells because the granules of these cells can be stained strongly with potassium bichromate.

———49. With regard to the **endocrine system**, which of the following statements is **true**?

- a. Somatostatin is secreted from the gamma cells of the pancreatic acini.
- b. Chromaffin cells which secrete catecholamines are found in the adrenal medulla.
- c. Aldosterone is secreted by the acidophilic cells of the pars distalis of the adenohypophysis.
- d. Tropic hormones are secreted from the pars tuberalis.
- e. Cortisol is secreted by the pancreas.

———50. **Hormones**,

- a) must be near their cell of origin to achieve maximum effect.
- b) must be present at high concentrations to elicit an effect.
- c) are produced mainly by exocrine glands.
- d) must consist of short-chain or long-chain polypeptides.
- e) have very specific effects on their target cells.

———51. Cells specialized to secrete **steroids** typically contain:

- a. extensive smooth endoplasmic reticulum.
- b. numerous mitochondria with tubular cristae.
- c. many lipid droplets.
- d. well-developed Golgi apparatus.
- e. all of the above.

———52. **Oxyphil cells**,

- a) are typically smaller than the parathyroid chief cells.
- b) contain abundant mitochondria.
- c) cytoplasm is characteristically basophilic.
- d) secrete calcitonin.
- e) are the predominant cell type in the parathyroid gland.

———53. **Epithelial cells** are arranged into cords around large capillaries or **sinusoids** in all of the following **EXCEPT** the:

- a. parathyroid.
- b. adenohypophysis.
- c. adrenal cortex.
- d. adrenal medulla.
- e. pancreatic islets.

———54. Which of the following statements is **false** in the **endocrine system**?

- a. Glucagon is secreted from alpha cells of the pancreas.
- b. Islets of Langerhans are most numerous in the tail of the pancreas.
- c. Prolactin is a protein hormone secreted by acidophilic lactotrophic cells.
- d. Infundibulum connects the pituitary to the hypothalamus.
- e. Median eminence is not part of the neurohypophysis.

———55. Select the correct statement concerning the **colloid within a thyroid follicle**.

- a) Is produced by the adenohypophysis and stored in the thyroid.
- b) Consists of numerous thyroglobulin-containing membrane-limited vesicles.
- c) Is pinocytosed and digested by follicular cells in response to increased circulating levels of thyroxine.
- d) Contains the storage forms of both T3 and T4.
- e) Normally lacks significant concentrations of iodine.

———56. **Testis** has the following coverings **except**:

- a. tunica vasculosa
- b. tunica albuginea
- c. tunica vaginalis
- d. tunica adventitia
- e. None

———57. In the **testis**, which statement is **true**?

- a. Interstitial cells (of Leydig) produce testicular fluid.
- b. The seminiferous tubules are lined with a complex stratified epithelium.
- c. The secondary spermatocytes are the largest cells of the spermatogenic lineage.
- d. The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
- e. During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.

———58. Which of the following is **NOT** considered part of the **male genital duct system**?

- a. Rete testis
- b. Tubuli recti
- c. Seminal vesicles
- d. Ductus deferens
- e. Ductus epididymidis

_____59. Which statement is **false** with regard to the **prostate gland**?

- a. It is the largest accessory structure of the male reproductive system.
- b. It is lined by simple columnar or pseudostratified epithelium.
- c. Corpora amylacea {prostatic concretions} are the spherical structures seen in some prostatic alveoli.
- d. It is composed of compound tubuloalveolar glands.
- e. It produces testicular androgens.

_____60. Which of the following statements is **false** in the **male reproductive system**?

- a. The secretion of seminal vesicle is particularly rich in fructose.
- b. Secondary spermatocytes are smaller than the primary spermatocytes.
- c. The epithelium lining the ductus epididymis is pseudostratified columnar.
- d. Tunica albuginea is the thickest in epididymis.
- e. The subcutaneous tissue in penis is devoid of fat.

_____61. The function of **Sertoli cells** includes all of the following **EXCEPT**:

- a. phagocytosis of residual bodies from maturing spermatids.
- b. establishment through tight junctions of a blood/testis barrier.
- c. secretion of testosterone.
- d. metabolic and mechanical support of germ cells.
- e. secretion of tubular fluid.

_____62. In the **male reproductive system**, which of the following statements is **false**?

- a) the seminiferous tubules are connected to the rete testis by the tubuli recti.
- b) interstitial cells of Leydig become functional at puberty due to the action of FSH.
- c) During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.
- d) The seminiferous tubules are lined with a complex stratified epithelium.
- e) Leydig cells produce testosterone.

_____63. During **spermiogenesis**, the **spermatid** does all of the following **EXCEPT**:

- a. acquire an acrosome.
- b. acquire a flagellum.
- c. condense the nuclear chromatin.
- d. shed the euchromatin into the residual body.
- e. shed excess cytoplasm into the residual body.

_____64. The **epithelial lining of the duct of the epididymis** is:

- a. simple squamous epithelium.
- b. ciliated columnar epithelium with goblet cells.
- c. transitional epithelium.
- d. pseudostratified epithelium with stereocilia.
- e. columnar epithelium with brush border and goblet cell

_____65. **Functions for the epididymis** include all of the following **EXCEPT**:

- a. provide a site for sperm maturation.
- b. secrete various factors into the tubular lumen.
- c. provide a site for sperm storage.
- d. absorb excess fluid entering from the testicular tubules.
- e. secrete the bulk of ejaculate volume.

_____66. The **seminal vesicle** consists of:

- a. serous secretory alveoli and branching ducts in a fibrous stroma.
- b. mucous secretory tubules.
- c. a coiled tubule lined by ciliated pseudostratified columnar epithelium.
- d. a hollow tube with an extensively folded mucosal lining, in a fibromuscular stroma.
- e. lymph nodules with crypts leading into the vas deferens.

_____67. The **ejaculatory duct** connects the:

- a. testis with the epididymis.
- b. epididymis with the seminal vesicle.
- c. seminal vesicle with the prostatic urethra.
- d. prostatic urethra with the bladder.
- e. prostatic urethra with the penile urethra.

———68. The **penile urethra** passes through the:

- a. spermatic cord. b. corpora cavernosa. c. corpus spongiosum. d. prostate. e. seminal vesicle.

———69. The spaces within erectile tissue of **corpora cavernosa** and **corpus spongiosum** are:

- a. vascular, lined by endothelium. b. lymphatic, lined by endothelium. c. interstitial, with no proper lining.
d. peritoneal, lined by mesothelium. e. urinary, lined by transitional epithelium.

———70. Which organ may **enlarge with age**, eventually causing **problems with urination**?

- a. testis b. epididymis c. vas deferens d. seminal vesicle e. prostate

———71. Which of the following is true of the **ampulla** of the **uterine tube** but **not** of its **isthmus**?

- a) Contains smooth muscle in its walls. b) Contains mucus-secreting peg cells.
c) Is covered by serosa derived from the peritoneum. d) Contains ciliated epithelial cells.
e) Contains large mucosal folds that are extensively branched.

———72. **Ovarian follicles** which are degenerating after failing to mature are called:

- a. primordial follicles. b. primary / secondary follicles. c. tertiary / Graafian follicles.
d. atretic follicles. e. corpora albicans.

———73. Which of the following statements is **true** about the **female reproductive system**?

- a. Mucosa of the uterus is commonly called perimetrium.
b. Fornix of vagina is lined by stratified squamous epithelium.
c. The life span of corpus luteum of pregnancy is about 10 lunar months.
d. Primary oocyte completes its first meiotic division at the time of fertilization.
e. Nonciliated secretory cells present in the lining epithelium of uterine tube synthesise and store glycogen.

———74. Which of the following statements is **true** in the **female reproductive system**?

- a. The cells of theca interna secrete progesterone.
b. Germinal (surface) epithelium covering the ovary gives rise to ova.
c. Glassy membrane is the last structure to disappear in an atretic follicle.
d. The alkaline environment of vagina prevents growth of microorganisms.
e. The basic pH in the vaginal lumen is due to the secretion of HCl by the tubuloalveolar glands of the vaginal walls.

———75. During **ovarian follicle maturation**, surrounding stromal cells differentiate to form the:

- a. granulosa. b. corona radiata. c. antrum. d. theca interna and theca externa. e. zona pellucida.

———76. After ovulation, **granulosa cells** and **cells of theca interna** grow and differentiate to form the:

- a. corpus albicans. b. corpus luteum. c. zona pellucida. d. corona radiata. e. cumulus oophorus.

———77. The **mucosal layer of the uterus** is called the:

- a. endometrium. b. myometrium. c. chorionic villi. d. placenta. e. cervix.

———78. The **proliferative phase of the endometrium** is characterized by all of the following **EXCEPT**:

- a. common mitotic figures in the endometrial stroma.
b. common mitotic figures in the glandular epithelium.
c. absence of differentiated secretory cells in the glandular epithelium.
d. relatively smooth epithelial lining of the glandular

tubules.

e. a stratified squamous surface.

———79. Which of the following statements is **false** with regard to the **circulatory system**?

- a) Lymphatic capillaries can penetrate the media of veins.
- b) Vasa vasorum are more frequent in veins than in arteries.
- c) The tunica adventitia tends to be much larger in veins than in arteries.
- d) The concentration of glycosaminoglycans is higher in arterial than in venous tissue.
- e) Sinusoids of the liver and the glomeruli of the kidney typically possess continuous endothelium.

———80. Which of the following statements is **true** with regard to the **circulatory system**?

- a. Continuous capillaries are common in brain and muscle tissue.
- b. Ventricles contain more abundant elastic fibers than do the atria.
- c. Tunical intima is homologous to the pericardium of the heart.
- d. The basal lamina of sinusoidal capillaries is continuous.
- e. The inner layer of a blood vessel wall, characterised by a simple squamous endothelium supported by a thin layer of connective tissue, is the tunica intima.

———81. With regard to the **white blood cells**, which of the following statements is **true**?

- a) Neutrophils are (normally) the most common white blood cells in circulating blood.
- b) A patient with a parasitic infection is likely to reveal an increase in the circulating monocytes.
- c) Basophils normally re-enter the bloodstream after leaving circulation and immigrating into peripheral tissues.
- d) A macrophage can originate from a mast cell.
- e) None of the above.

———82. Which of the following statements is false with regard to the circulatory system?

- a. **Post-capillary venules lack a smooth muscle media.**
- b. **Carotid bodies are chemoreceptors sensitive to hypoxia.**
- c. **The brachiocephalic and subclavian arteries are elastic arteries.**
- d. **Continuous capillaries often have phagocytic cells inserted between the endothelial cells of their lining.**
- e. **A prominent inner elastic membrane (internal elastic lamina), often appearing in cross section as a wavy line, is characteristic of muscular arteries..**

———83. Which of the following statements is true with regard to the circulatory system?

- a. **Albumin is present in higher concentrations in plasma than in serum.**

b. Smooth muscle is most substantial in the media of muscular arteries.

c. Heart valves normally consist of an endothelial surface covering loose areolar connective tissue.

d. Pericytes, which are terminally differentiated and multinucleated, are specialized cardiac muscle cells.

e. The SA node, the AV node, and the Purkinje fibers of the myocardium all consist of specialized nerve cells.

———84. Which of the following statements is false with regard to the histology of the heart?

a. Ventricles contain more abundant elastic fibers than do the atria.

b. Adipose tissue that generally surrounds the heart accumulates in the epicardium.

c. The coronary arteries branch to supply the epicardium, myocardium, and endocardium,

d. Endocardium is the thickest in atria and the thinnest in ventricles, especially the left ventricle.

e. Cardiac muscle is more vascularized and has more abundant mitochondria than does skeletal muscle.

——85. Which of the following is partly encapsulated and covered by nonkeratinized stratified squamous epithelium?

a) palatine tonsil b) lingual tonsil c) appendix d) Peyer's patch e) pharyngeal tonsil

———86. Which of the following lymphoid organs receives afferent lymphatic vessels?

a) appendix b) palatine tonsil c) spleen d) thymus e) lymph node

———87. Thymus-dependent region of a peripheral lymphoid organs include,

- a) medullary cords of lymph nodes.
- b) periarterial lymphatic sheaths of the spleen.
- c) Hassall's corpuscles of the thymus.
- d) deep crypts of the pharyngeal tonsils.
- e) red pulp cords (Billroth's cords) of the spleen.

———88. The spleen:

- a) Has smooth myocytes in its capsule.
- b) Filters lymph.
- c) Has no B lymphocytes.
- d) Has an outer cortex and inner medulla.
- e) Contains periarterial lymphoid sheath in its red pulp.

———89. Regarding the thymus:

- a) Has lymphoid nodules in its cortex.
- b) Is relatively more developed with increasing age.
- c) among blood vessels, contains only sinusoids.
- d) Has blood-thymus barrier in the cortex.
- e) Contains more B lymphocytes than the T lymphocytes.

———90. Which of the following components of the respiratory tract wall increase(s) in amount from trachea to alveoli?

- a) cilia
- b) smooth muscle
- c) elastic fibers
- d) cartilage
- e) goblet cells

———91. Which of the following structures is (are) partly lined by nonkeratinized stratified squamous epithelium?

- a) nasopharynx
- b) larynx
- c) bronchi
- d) trachea
- e) bronchioles

———92. In which of the following are goblet cells most frequent?

- a. trachea
- b. bronchus
- c. bronchiole
- d. alveolus

e. The frequency of goblet cells in each of the above is about equal.

—————93. As respiratory passages branch from trachea to bronchioles, the epithelium gradually changes from:
a. simple squamous to ciliated, pseudostratified columnar.
b. simple columnar to simple squamous.
c. simple cuboidal to simple columnar. d.
stratified squamous to simple squamous. e. ciliated. pseudostratified columnar to simple cuboidal.

—————94. Alveolar type I cells are:
a. cuboidal cells that secrete surfactant. b. squamous cells involved in gas exchange. c. ciliated cells that move mucus. d. columnar cells that secrete proteolytic enzymes. e. ameboid and phagocytic cells that clean the alveolar surface.

—————95. Bronchioles:
a) Have well developed hyaline and elastic cartilages.
b) Are lined by stratified squamous epithelium.
c) Have no muscle. d) Have Clara cells. e)
Have mucous producing glands.

—————96. Pulmonary surfactant,
a) prevents alveolar collapse by increasing surface tension.
b) is secreted by type I pneumocytes.
c) consists of a monolayer of phospholipids overlying an aqueous hypophase. d) may promote the growth of bacteria.
e) excessive production in the lungs of premature infants causes respiratory distress syndrome.

—————97. Nasal mucosa includes:
a. pseudostratified, ciliated columnar epithelium. b. lamina propria. c. seromucous glands.
d. venus plexuses. e. all of the above.

———98. Which of the following is lined by a **mucosa**?

- a. Peritoneal cavity b. Pericardial cavity c. Pleural cavity d. Alimentary canal e. All of them are lined by a mucosa

———99. Voluntary neural control of **urine flow in males** is managed by

- a) neural control of ADH release by the hypothalamus. b) neural control of aldosterone production by the adrenal glands.
c) regulation of glomerular filtration rate through innervation of the afferent arterioles.
d) innervation of an internal sphincter around the prostatic urethra as it exits the bladder.
e) innervation of the external sphincter around the membranous urethra as it exits the pelvic cavity.

———100. With regard to the **tonsils**, which statement is **true**?

- a) Palatine tonsils are lined with a squamous keratinized stratified epithelium without crypts.
b) Pharyngeal tonsil is covered by transitional epithelium and contains diffuse lymphoid tissue and lymphoid nodules.
c) Pharyngeal tonsil has no crypts, and its capsule is thinner than the capsule of the palatine tonsils.
d) The lingual tonsils are smaller and more numerous than the palatine and pharyngeal tonsils.
e) Each lingual tonsil has a single crypt and is supported by a reticulum of epithelial cells of endodermal origin.